



The Animal We Still Are

Instinct, belonging, territory, cooperation and culture: what remains of the animal beneath human intelligence.

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We can build laws, cities, religions, armies and theories about ourselves. But there are moments when the body acts before intelligence has time to explain who we believe we are.

Some time ago—years ago now—I was carrying my barely two-year-old son on my shoulders. I stepped badly, stumbled and fell. Everything happened in a fraction of a second. The most useful reaction for protecting my own body would have been to let go of him and put out my hands to break the fall. I did exactly the opposite. I held on to him, lifted him above my head so he would not hit the ground, and ended up cushioning his fall with my arms and my own body. He barely felt the impact. I took quite a hit.

The interesting part came later, when I thought about it. I do not remember making any decision. I did not think that I would rather hurt myself than let him get hurt. I did not assess the risk, calculate the trajectory of the fall or consciously decide to sacrifice myself. There was no time. My body simply made protecting him the priority.

That is why I do not consider it an act of courage either. Courage requires some awareness of what one is doing: perceiving the danger, understanding the risk and acting despite it. In that moment, the reaction came first. It was faster than any conscious reasoning and, probably, much older.

That small accident contains a fascinating part of what we are. We like to define ourselves by our intelligence, because it is one of the traits that most clearly distinguishes us from other animals and the one that allowed us to develop complex language, cumulative culture, technology and civilisation. But that intelligence did not appear by replacing the animal we once were. It was built on top of it.

We still protect our young, seek belonging in a group, defend spaces, respond to threats, compete for resources, form bonds, recognise hierarchies, cooperate and distinguish with remarkable ease between those we consider our own and those who remain outside. Then our intelligence transforms those predispositions into far more complex behaviours: a house replaces a shelter, a property deed formalises a territory, a flag represents millions of individuals, a uniform turns strangers into comrades, and a law can force us to restrain a response that our body had already begun to prepare.

Recognising this does not make us less rational.

It makes us more difficult to understand.

Because a large part of human history can be viewed as the history of an extraordinarily intelligent animal that learned to build upon mechanisms far older than its own species.

What Acts Before Reason

Instinct, Reflex and Predisposition

Before going any further, it is worth pausing over a word we use far too easily: instinct. In everyday language, we call almost any behaviour that seems to arise without reflection instinctive. In biology and ethology, the issue is considerably more complicated. A reflex, an automatic response, an emotion, an inherited predisposition and a learned behaviour are not the same thing, even though from the outside they may look similar.

Pulling your hand away from a burning surface is a relatively simple example. Part of the response can be organised through spinal-cord circuits before the conscious experience of pain has fully formed. Protecting a child, defending a territory or cooperating with a group, by contrast, are enormously more complex behaviours. Biological predispositions, hormonal and neural systems, experience, learning and social context all play a role. In our species, we must also add a culture capable of profoundly altering how all those elements are expressed.

So when we speak here of human instincts, we should not imagine rigid instructions written into our genes. A gene does not contain the order “protect your child”, “defend your home” or “support your team”. Genes participate in building and regulating organisms whose nervous systems make certain tendencies and responses possible, and those tendencies then develop within an environment. The relationship between genes and behaviour is far more complex than a chain of commands.

Evolution does not require intention either. A DNA molecule does not want to perpetuate itself. But if certain variants help produce organisms that leave more copies of those variants in subsequent generations, their frequency can increase. After millions of years, the result can appear strikingly intentional even though there is no intention behind it.

My body did not need to understand natural selection in order to try to protect my son.

The Offspring Before Oneself

Parental protection is one of the best places to observe this logic because it forces us to abandon an overly simple view of evolution: the idea that every organism should always try to save itself first.

In many species, exactly the opposite occurs. Birds attack animals far larger than themselves when they approach the nest; mammals place themselves between a predator and their young; parents devote enormous amounts of energy to feeding offspring that cannot yet give anything back. In some species, reproducing and caring for offspring even causes a drastic reduction in the parent's own chances of survival.

This stops seeming paradoxical once we change the scale. Natural selection does not reward longevity for its own sake. A trait that allows an organism to live a long time but leaves no descendants disappears with the individual that carries it. Another trait that involves taking certain risks but sufficiently increases the survival of descendants who share part of that inheritance can spread.

In the 1960s, William D. Hamilton formalised a fundamental part of this idea through the theory of kin selection and the concept of inclusive fitness. Helping another individual can have different evolutionary consequences depending on relatedness and on the costs and benefits of the behaviour. A father does not unconsciously solve an equation before saving his child; the equation belongs to the model we use to explain why organisms capable of reacting in that way could have evolved.

And kinship does not fully explain human beings.

The Animal That Needs Others

A human being alone is a fairly helpless animal. We are born extraordinarily dependent and remain so for years. For most of our history, survival meant belonging to groups capable of finding food, defending themselves, raising children, transmitting information and caring for individuals who were temporarily unable to fend for themselves. We are gregarious animals, just as a sardine is within a shoal, a wildebeest within a herd or a dog within its group. The form that organisation takes differs in every species, but the basic need to belong to a collective that increases the chances of survival is not uniquely ours.

Our sociability is not exceptional simply because it exists. There are animal societies far more integrated than our own. Ants, bees and termites take cooperation to extremes in which enormous numbers of individuals do not reproduce at all. Fish form shoals, birds form flocks, many herbivores form herds, and numerous primates maintain complex social relationships involving alliances, competition, reconciliation, kinship and hierarchies.

What is distinctive about human beings lies elsewhere. Our intelligence and symbolic capacity allow us to construct group belonging from categories, stories, symbols or shared ideas. We do not need to know every member personally. We can regard someone as one of “us” simply because they share a flag, a language, a religion, a team, a nationality or a particular collective identity.

Family. Village. Religion. Nation. Political party. Company. Football team.

Any of them can become “us”.

Us and Them

We Have Won

A stadium lets us observe this in an almost experimental form. Thousands of people gather to watch a small group of individuals try to put a ball into a goal more times than another group. From a purely material point of view, the result may change absolutely nothing in the spectator’s life. Yet the heart speeds up, tension appears, people shout, strangers embrace one another, and a victory can produce euphoria that lasts for hours.

And we use a particularly revealing expression: “we have won”.

I have not run. I have not touched the ball. I have not trained with the team and I probably do not personally know any of the players. Yet I do not say “they won”. I say “we won”. The group has incorporated its success into my own identity.

That does not turn football into a simple animal behaviour. It is an extraordinarily sophisticated cultural product, with rules, institutions, contracts, markets, traditions and an enormous symbolic weight. But it uses a much older capacity: our ability to identify with a group and experience part of what happens to it as though it were happening to us as well.

That is why it is not surprising that, under certain circumstances, the reverse can also appear. The rival can stop being merely the person playing against our team and become “them”. Colours make that distinction immediately visible. The shirt, crest and scarf signal belonging. Most people keep all of this within the game, but for a minority rivalry can turn into hostility and even violence.

Intelligence invented football.

It did not invent “us”.

Our Own and the Others

Social psychology has spent decades showing how little we need in order to form groups. In Henri Tajfel’s classic experiments on the so-called minimal group paradigm, assigning people to almost arbitrary categories was enough for certain preferences toward members of their own group to emerge.

That does not mean human beings are condemned to hate strangers. Favouring one’s own group and hostility toward outsiders are not exactly the same phenomenon. We can favour our own without attacking anyone, and we can greatly expand what we consider to be “ours”.

But the ease with which we construct social categories is difficult to ignore.

It also helps explain an everyday situation. We can have far more in common with someone outside our group than with someone inside it. Two supporters of rival teams may share a profession, musical tastes, political ideas, age, friends and virtually all of their important interests. For ninety minutes, however, a completely different characteristic can take priority.

In much more extreme circumstances, something similar happens with a border.

The Flag and the Territory

A dog can leave scent marks in a territory. We plant a flag.

The comparison has obvious limits. A flag is not an animal territorial marker directly transformed by evolution. For it to exist, language, symbolism, history and shared conventions are required. But both behaviours allow us to ask an interesting question: what did our intelligence do with predispositions related to space, resources and what we consider our own?

We are territorial animals too. Like many primates, dogs or lions, we distinguish spaces we feel are ours from others we regard as belonging to someone else, and we react when we perceive an intrusion. Not all gregarious species work this way: a sardine can belong to an enormous shoal without defending a stable territory. We combine both tendencies. We are gregarious and territorial.

Our home has walls, locks and property deeds. Our country has borders that may cross mountains and rivers we have never seen. We can feel that a territory thousands of kilometres away is ours and become emotional at the sight of a piece of cloth that represents it.

The difference, once again, lies in what our intelligence builds on top. An animal can defend a concrete area. We can turn that same notion of territory into a house, an estate, a border, a nation or a map.

We are also highly visual primates. While other species depend far more on smell to recognise individuals, reproductive states or territories, we have turned colours and shapes into a fundamental part of our social universe. Uniforms, flags, insignia, coats of arms, crowns, signs, brands and clothing allow categories to be recognised almost instantly.

Physical territory eventually became symbolic territory as well.

We do not only defend our home.

We defend “what is ours”.

The Stranger Who Could Have Been a Friend

War takes this capacity to one of its most terrible extremes. Two soldiers may face one another without knowing each other. Perhaps they are the same age, have young children, similar jobs and almost identical tastes. They might listen to the same music, enjoy the same books and laugh at the same things. Had they met in a bar in another country, they might have ended up talking for hours.

But they are wearing different uniforms.

One is one of ours and the other one of theirs.

At the same time, a soldier may risk his life for a comrade with whom he shares very few personal affinities. That comrade may have values, ideas or behaviours profoundly different from his own, even some he would completely reject outside that context. And yet, in that moment, he belongs to “us”. The other soldier, perhaps far more similar in tastes, character or principles, is wearing another uniform.

There are even documented cases of soldiers covering a grenade with their own bodies to protect those around them. There is no need to reduce such an extreme act to a single “group instinct”, but neither should we ignore how powerfully belonging can override other differences. Training, bonds, identity, norms and culture can reinforce a much older predisposition: protecting those we recognise as part of our group, even when doing so means assuming an enormous risk to ourselves.

Sometimes a colour, an insignia or a flag is enough to decide, at a surprisingly primary level, who belongs inside the group we protect and who remains outside. We can include unrelated individuals, even people very different from us, within a social category whose protection can come to compete with our own survival.

And our intelligence can take that capacity in two opposite directions. It can expand “us” to include millions of strangers, or shrink it until someone who might otherwise seem extraordinarily close to us becomes an enemy simply because of a symbolic boundary.

Cooperation, Law and Hierarchy

Cooperating with Strangers

Here human beings differ from many forms of animal organisation in the scale and flexibility of our cooperation.

Every morning, millions of people get into cars built by people they will never meet, travel on roads constructed with taxes paid by strangers, obey signs designed by others, buy food grown hundreds of kilometres away and use money whose value depends on an enormous community accepting the same convention.

Almost none of these exchanges requires kinship or friendship.

Human cooperation is sustained by an extraordinary combination of reciprocity, reputation, norms, trust, punishment, institutions and culture. We can cooperate with someone because we expect to see them again, because others will know how we behaved, because there is a rule we consider fair, or simply because we have learned that certain forms of cooperation are right.

Our intelligence made it possible to build gigantic societies on social capacities that evolved in much smaller groups.

But as the group grew larger, another problem grew with it.

How do we stop each individual from acting only according to their immediate interests?

The Law Against Impulse

One answer was law.

One of the most extraordinary features of our societies is that we have created explicit rules capable—at least ideally—of constraining both the person who feels an impulse and the person strong enough to act on it.

If someone hits me, I may feel an immediate urge to hit back. Society sets limits on what I can do. If someone has something I need, I cannot simply take it from them. If a person belongs to another group, they still retain rights. If someone commits a crime, even the victim must accept that a collective system will determine the response.

We do not need to assume that each of these tendencies is a separate instinct. It is enough to recognise that human beings have emotions, interests, group preferences, potential aggression and responses to threats, and that a society made up entirely of individuals who always obeyed their immediate reactions would have enormous difficulty surviving.

Laws do not prove that we have left our animal nature behind. In a sense, they show that we are capable of recognising it and building mechanisms to limit some of its consequences.

Culture does not necessarily eliminate a predisposition.

It can teach us not to obey it.

Hierarchy and Position

Something similar happens with status. Many social species organise relationships of dominance, access to resources, reproductive priority or alliances. Their structures are so different that speaking of a single “animal hierarchy” would make little sense, but attention to relative position within the group appears repeatedly in nature.

Human beings are no exception. We know who runs an organisation, who has more resources, who has prestige, who can make decisions and who receives more attention. Even in communities that explicitly defend equality, forms of reputation and recognition emerge.

Again, the difference lies in what we build on top. We can inherit titles, elect presidents, award prizes, accumulate followers on a social network or measure prestige with numbers that represent no immediate physical resource. We have turned status into another abstraction.

And then we built institutions precisely to limit it. A constitution can establish that the person with the greatest political power cannot do certain things. A law can grant the same right to someone at the bottom of the social ladder as to someone at the top.

Our intelligence did not eliminate sensitivity to hierarchy.

It invented ways to question it.

The Animal in Everyday Life

The Dog on the Other Side of the Door

There is one species in which this combination of sociability, cooperation and territory is especially visible to us because it has lived beside us for thousands of years.

The dog.

Humans and dogs come from very different lineages, but both are social animals. Over thousands of years, domestication selected animals increasingly capable of living alongside us, and today's dogs show an extraordinary sensitivity to certain human signals. They follow pointing gestures, seek information in our faces, form attachment bonds and learn the social rules of a household.

They also understand perfectly well that spaces and groups exist. A dog may calmly accept everyone who belongs to its familiar environment and react differently to a stranger who crosses a doorway.

We make a similar distinction, although we have surrounded it with locks, doorbells, property deeds, alarms, doormen and laws against trespassing.

Part of the extraordinary ease with which the two species have learned to live together probably comes from the fact that we share two very old characteristics: we are gregarious animals and we manage territories. For a dog, it makes sense to distinguish who belongs to the group and who does not, what space forms part of its familiar environment, whom it should protect and when the appearance of a stranger may require attention or defence.

And perhaps, at a much more primary level than we like to admit, we do exactly the same.

We know who is one of ours and who is not. We know which space is ours and which feels foreign. We react when someone enters a home, threatens a member of the group or crosses a boundary we consider our own. The difference lies in everything our intelligence has built on top of that initial impulse.

The dog can mark territory with scent. We use a lock, a property deed, a flag or a border. The dog recognises those who belong to its group through coexistence, smell, voice and experience. We add surnames, uniforms, documents, symbols, anthems and shared stories. But when we react to an invasion of our space or to a threat against someone we regard as one of our own, the response can begin at a level as fast and as little deliberative as that moment when, as I fell, I held my son before I had time to decide what I ought to do.

It is not that our intelligence replaced those mechanisms with more sophisticated ones. It dressed them.

It turned territory into property, group into identity, belonging into symbol and defence into norm. Then it built arguments capable of explaining, justifying, limiting or even contradicting those responses.

But often the reaction comes first.

Perhaps part of the extraordinary success of our relationship with dogs comes precisely from there. Two social species capable of recognising group, territory, belonging and threat found a deep compatibility through domestication. They learned to read us, and we learned to read them.

We do not think alike.

But sometimes we react from surprisingly similar places.

Fear, Disgust and Survival

Not all of our ancient systems concern other individuals. Some protect the organism directly.

Fear prepares the body for a possible threat. Heart rate can change, attention narrows and the organism mobilises resources without our consciously ordering each response. Then our intelligence can use those same systems to fear things no distant ancestor could have imagined: losing a job, failing an exam, suffering an economic crisis or receiving tomorrow a piece of news that does not yet exist.

Disgust offers another example. Rotten food, certain bodily secretions, faeces or corpses can provoke rejection before we perform any microbiological analysis. It has been proposed that some of these responses form part of a pathogen-avoidance system. Other animals also show behaviours aimed at avoiding contaminated food, sick individuals or places associated with health risks.

Once again, we built on top of it.

From aversion to contamination, we moved on to hygiene rules, sewer systems, food preservation, sterilisation and medicine. A response that may have very ancient roots ends up connected to a modern laboratory.

Culture, Reason and Nature

Culture Changes Us Too

It would be a mistake to stop here and conclude that beneath a thin cultural layer there is a "true animal" that determines what we do. Culture is not decoration placed on top of biology.

It is part of us too.

From birth, we learn which behaviours are acceptable, how much distance to keep from another person, what foods we can eat, which parts of the body we

show, how we express anger, whom we call family and which symbols we regard as our own. Different societies can produce very different responses from similar biological predispositions.

Cultural evolution is also far faster than genetic evolution. A norm can change in a generation. A law can change in an afternoon. A technology can alter within a few years behaviours that remained relatively stable for centuries.

Biology and culture are not adversaries. They have been changing one another for thousands of years.

Even our capacity to learn culture is a product of evolution.

Natural Does Not Necessarily Mean Good

Recognising a biological root in a behaviour does not justify it either.

This point is essential.

If aggression has evolutionary components, an act of aggression does not become morally right. If we readily favour our own group, discrimination is not justified. If jealousy, competition or territoriality exist, none of those predispositions determines how a society ought to be organised.

Confusing what exists in nature with what ought to be morally acceptable is known as the naturalistic fallacy.

Precisely because we can evaluate our own responses, we can do something extraordinary: act against them.

I can feel anger and not strike. I can feel fear and move forward. I can spontaneously prefer someone from my own group and decide to treat someone outside it impartially. I can desire revenge and accept justice.

Intelligence can find better ways to satisfy our impulses and, to some extent, allow us to decide which ones we do not want to obey.

Reason Appeared Inside the Animal

For a long time, we have represented human evolution almost as a ladder. First the animal, then the primate, then the human being and finally reason, as though each rung had replaced the previous one.

Evolution does not work that way.

It works with what already exists. It modifies structures, reuses circuits and builds new capacities on earlier systems. Our brain retains structures and functions with very different evolutionary histories, deeply interconnected with one another. There is no small animal brain hidden beneath a rational one, but there is an enormous biological continuity.

Reason appeared inside an organism that already felt fear, sought food, reproduced, protected offspring and needed relationships with others.

Then it learned to observe those same tendencies.

Perhaps that is why we can rationally understand that the result of a football match has no important consequence for our survival and, seconds later, jump off the sofa shouting at a goal. We can know that a flag is cloth and still be moved by it. We can defend equality and discover that our brain spontaneously classifies people.

And I can know perfectly well that when you fall, you should put out your hands.

But that time I was carrying my son on my shoulders.

And my hands did something else.

The Animal That Built on Top of Itself

Civilisation emerged as one consequence of an animal developing extraordinarily flexible intelligence and an exceptional capacity to accumulate culture.

Social animals built cities. Animals capable of competition designed markets and rules. Animals sensitive to status invented institutions to distribute and limit power. Animals capable of aggression wrote laws forbidding its unrestricted use. Animals that protected their young built schools and hospitals. Animals that avoided disease developed medicine. Territorial animals drew borders and then signed treaties to decide how to cross them.

And an animal that distinguished its own from strangers eventually formulated an extraordinary idea: that certain rights should belong to every human being precisely regardless of the group to which they belong.

Perhaps some of the greatest challenges and possibilities opened by our intelligence lie there: understanding our nature well enough to decide what we want to do with it.

What We Still Are

Just Another Animal

We have grown accustomed to saying “human beings and animals” as though we were talking about two separate categories.

We never stopped belonging to the second.

We still compete for resources, seek shelter, protect our young, form groups, defend spaces, cooperate, fear, desire, play and learn. We share many of these needs with animals enormously distant from us and others with our closest evolutionary relatives.

All these impulses can also be understood as part of the way evolution found, in our species, to favour the continuity of our biological inheritance. The molecule does not want to survive, think or decide anything, but variants that helped produce organisms capable of protecting themselves, protecting their offspring, cooperating with others or defending resources had a better chance of continuing. In other species, the same logic has produced different strategies. In ours, it produced, among many other things, these mechanisms that still shape a deep part of our behaviour today.

Our difference lies in the means made available to us by a particular combination of natural factors: intelligence, language, cooperation and cumulative culture.

With them, we have occupied virtually the entire planet. We have transformed ecosystems, domesticated other species, built cities visible from space, modified organisms and created machines capable of performing cognitive tasks that for a long time we considered exclusively ours.

We have even begun to imagine intelligences that may one day no longer need a biological body.

But we came from one.

What Would Remain of Us

And this brings the question back to the world of *EIDOS*.

If one day we could transfer a human mind to another substrate, what would we have to preserve in order to claim that it remained human?

Memory seems obvious. So do personality, knowledge, the capacity to reason and individual history. But what would happen to everything we never consider when we list who we are?

Fear.

Attachment.

Desire.

The need to belong.

The response to a threat.

The protection of one's children.

The satisfaction of being accepted by a group.

The unease caused by a stranger.

The reaction that makes hands protect a child before there is time to decide to do so.

EIDOS also presents a particular problem. Many of these mechanisms developed in organisms subject to a condition that seemed unavoidable: death. Protecting oneself, reproducing, caring for offspring, cooperating with the group or competing for resources makes sense within a limited existence, in which the individual disappears and continuity depends, in one way or another, on what is left behind.

But what happens if we remove that condition while preserving the mind that formed under it? Individuals may stop dying without attachment, territoriality, the need to belong, fear of loss, competition or many other mechanisms shaped over millions of years in mortal organisms necessarily disappearing. Society is no longer subject to one of the fundamental rules that shaped those impulses, yet it remains inhabited by minds built under that rule.

That is one of the conflicts in *EIDOS*. Eliminating death would not erase what a history of mortality left inside us.

If we removed everything we share with other animals, perhaps we would not obtain a purer and more rational version of the human being. Perhaps we would have removed precisely part of what we were trying to preserve.

The Same Space

Intelligence has given us an extraordinary advantage over virtually every other species. We do not run especially fast, we have no large fangs, we cannot fly, we cannot breathe underwater and our senses are mediocre compared with those of many animals. But we build cars, weapons, aircraft, submarines, telescopes and sensors.

Our intelligence expanded our means to a point no other known species has reached.

That can make us forget that we are still here with them. Sharing the same planet, the same resources and an evolutionary history that, if traced far enough back, eventually converges.

We are not visitors to nature or spectators standing outside it. We are one of its outcomes. An animal that came to understand natural selection, sequence the DNA from which it is built and wonder why it protects its child while falling.

Intelligence allowed us to understand part of our nature. Culture allowed us to transform it. Laws help us restrain it, and ethics may allow us to decide what to do with it.

But none of those capacities requires us to deny where we came from.

After all, we are just another animal competing, cooperating and trying to survive in the same space as all the others. We have far more means because our own nature allowed us to build them, but those means do not replace the primary mechanisms on which they were built. We remain capable of reacting to offspring, group, territory, threat or belonging before we have turned them into arguments. Laws, culture and ethics can restrain those responses, set limits on them and even force us to act against them; that is precisely why we built them. They do not erase the animal. They prevent it from always having the last word.

We did not stop being animals when we learned to think.

We learned to think while being animals.

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